ac 负载分担双链路热备份

一、实验目的

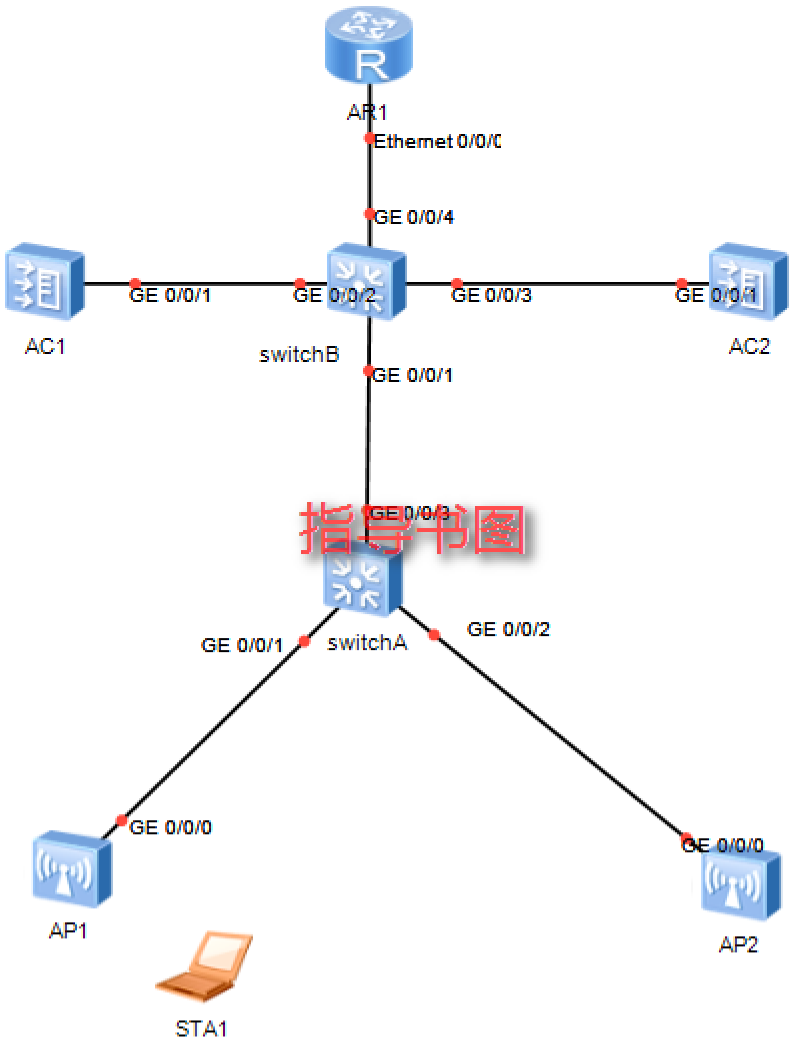
了解 ac 负载分担双链路热备份相关知识、目的、配置方法。

二、模拟器版本

130

三、实验拓扑

3.1 网络拓扑



3.2 ip 规划表

配置项	规划数据
管理 VLAN	VLAN100
STA 业务 VLAN	VLAN 101
AC 备用 VLAN	VLAN 102
DHCP 服务器	Router 作为 DHCP 服务器，为 AP 和 STA 分配地址，
	STA 默认网关为 10.23.101.1/24 AP 默认网关为 10.23.100.1/24
AP 的 IP 地址池	10.23.100.4 ~ 10.23.100.254/24

STA 的 IP 地址池	10.23.101.2 ~ 10.23.101.254/24	
AC 的源接口	VLANIF100：10.23.100.1/24	
AC1 的管理 IP 地址	VLANIF100：10.23.100.2/24	
AC2 的管理 IP 地址	VLANIF100：10.23.100.3/24	
主用 AC 和备用 AC	AC1 作为 AP1 的主用 AC、AP2 的备用 AC AC2 作为 AP2 的主用 AC、AP1 的备用 AC	
AC1 的主备通道 IP 地址和端口号	IP 地址：VLANIF102，10.23.102.1/24	
	端口号：10241	
AC2 的主备通道 IP 地址和端口号	IP 地址：VLANIF102，10.23.102.2/24	
	端口号：10241	
AP 组	名称：ap1	
	引用模板：VAP 模板 wlan-net、域管理模板 default、 AP 系统模板 ap-system1	
	名称：ap2	
	引用模板：VAP 模板 wlan-net、域管理模板 default、 AP 系统模板 ap-system2	
AP 系统模板	名称：ap-system1 主用 AC：AC1 备用 AC：AC2 名称：ap-system2 主用 AC：AC2 备用 AC：AC1	
域管理模板	名称：default 国家码：中国	
SSID 模板	名称：wlan-net SSID 名称：wlan-net	
安全模板	名称：wlan-net	
	安全策略：WPA-WPA2+PSK+AES	
	密码：a1234567	
VAP 模板	名称：wlan-net 转发模式：直接转发 业务 VLAN：VLAN 101	
	引用模板：SSID 模板 wlan-net、安全模板 wlan-net	

四、实验步骤

4.1 根据网络拓扑在模拟器中搭建网络;

4.2 步骤 1 配置周边设备

配置 SwitchA 连接 AP 的接口 GE0/0/1 和 GE0/0/2 的 PVID 为 VLAN100 并加入 VLAN100 和 VLAN101，SwitchA 连接 SwitchB 的接口 GE0/0/3 加入 VLAN100 和 VLAN101。

```
<HUAWEI> system-view
[HUAWEI] sysname SwitchA
[SwitchA] vlan batch 100 101
[SwitchA] interface gigabitethernet 0/0/1
[SwitchA-GigabitEthernet0/0/1] port link-type trunk
[SwitchA-GigabitEthernet0/0/1] port trunk pvid vlan 100
[SwitchA-GigabitEthernet0/0/1] port trunk allow-pass vlan 100 101
[SwitchA-GigabitEthernet0/0/1] port-isolate enable
[SwitchA-GigabitEthernet0/0/1] quit
[SwitchA] interface gigabitethernet 0/0/2
[SwitchA-GigabitEthernet0/0/2] port link-type trunk
[SwitchA-GigabitEthernet0/0/2] port trunk pvid vlan 100
[SwitchA-GigabitEthernet0/0/2] port trunk allow-pass vlan 100 101
```

```
[SwitchA-GigabitEthernet0/0/2] quit
[SwitchA] interface gigabitethernet 0/0/3
[SwitchA-GigabitEthernet0/0/3] port link-type trunk
[SwitchA-GigabitEthernet0/0/3] port trunk allow-pass vlan 100 101
[SwitchA-GigabitEthernet0/0/3] quit
```

配置汇聚交换机 SwitchB 连接 SwitchA 的接口 GE0/0/1 加入 VLAN100 和 LAN101，SwitchB 连接 AC1 的接口 GE0/0/2 和 SwitchB 连接 AC2 的接口 GE0/0/3 加入 LAN100 和 VLAN102，SwitchB 连接 Router 的接口 GE0/0/4 加入 VLAN100 和 VLAN101。

```
<HUAWEI> system-view
[HUAWEI] sysname SwitchB
[SwitchB] vlan batch 100 to 102
[SwitchB] interface gigabitethernet 0/0/1
[SwitchB-GigabitEthernet0/0/1] port link-type trunk
[SwitchB-GigabitEthernet0/0/1] port trunk allow-pass vlan 100 101
[SwitchB-GigabitEthernet0/0/1] quit
[SwitchB] interface gigabitethernet 0/0/2
[SwitchB-GigabitEthernet0/0/2] port link-type trunk
[SwitchB-GigabitEthernet0/0/2] port trunk allow-pass vlan 100 102
[SwitchB-GigabitEthernet0/0/2] quit
[SwitchB] interface gigabitethernet 0/0/3
[SwitchB-GigabitEthernet0/0/3] port link-type trunk
[SwitchB-GigabitEthernet0/0/3] port trunk allow-pass vlan 100 102
[SwitchB-GigabitEthernet0/0/3] quit
[SwitchB] interface gigabitethernet 0/0/4
[SwitchB-GigabitEthernet0/0/4] port link-type trunk
[SwitchB-GigabitEthernet0/0/4] port trunk allow-pass vlan 100 101
[SwitchB-GigabitEthernet0/0/4] quit
```

配置 Router 连接 SwitchB 的接口 GE0/0/1 加入 VLAN100 和 VLAN101。

```
<Huawei> system-view
[Huawei] sysname Router
[Router] vlan batch 100 101
[Router] interface gigabitethernet 0/0/1
[Router-GigabitEthernet0/0/1] port link-type trunk
[Router-GigabitEthernet0/0/1] port trunk allow-pass vlan 100 101
[Router-GigabitEthernet0/0/1] quit
```

4.3 步骤 2 配置 AC 与其它网络设备互通

配置 AC 的接口 GE0/0/1 加入 VLAN100 和 VLAN102。

```
<AC6605> system-view
[AC6605] sysname AC1
[AC1] vlan batch 100 to 102
[AC1] interface vlanif 100
[AC1-Vlanif100] ip address 10.23.100.2 24
[AC1-Vlanif100] quit
[AC1] interface vlanif 102
[AC1-Vlanif102] ip address 10.23.102.1 24
[AC1-Vlanif102] quit
[AC1] interface gigabitethernet 0/0/1
```

[AC1-GigabitEthernet0/0/1] port link-type trunk
[AC1-GigabitEthernet0/0/1] port trunk allow-pass vlan 100 102
[AC1-GigabitEthernet0/0/1] quit
配置 AC2 的接口 GE0/0/1 加入 VLAN100 和 VLAN102。
<AC6605> system-view
[AC6605] sysname AC2
[AC2] vlan batch 100 to 102
[AC2] interface vlanif 100
[AC2-Vlanif100] ip address 10.23.100.3 24
[AC2-Vlanif100] quit
[AC2] interface vlanif 102
[AC2-Vlanif102] ip address 10.23.102.2 24
[AC2-Vlanif102] quit
[AC2] interface gigabitethernet 0/0/1
[AC2-GigabitEthernet0/0/1] port link-type trunk
[AC2-GigabitEthernet0/0/1] port trunk allow-pass vlan 100 102
[AC2-GigabitEthernet0/0/1] quit

4.4 步骤 3 配置 Router 作为 DHCP 服务器为 STA 和 AP 分配 IP 地址

在 AC 上配置 VLANIF100 接口为 AP 提供 IP 地址。
[Router] dhcp enable
[Router] ip pool sta
[Router-ip-pool-sta] network 10.23.101.0 mask 24
[Router-ip-pool-sta] gateway-list 10.23.101.1
[Router-ip-pool-sta] quit
[Router] ip pool ap
[Router-ip-pool-ap] network 10.23.100.0 mask 24
[Router-ip-pool-ap] excluded-ip-address 10.23.100.2
[Router-ip-pool-ap] excluded-ip-address 10.23.100.3
[Router-ip-pool-ap] gateway-list 10.23.100.1
[Router-ip-pool-ap] quit
[Router] interface vlanif 100
[Router-Vlanif100] ip address 10.23.100.1 24
[Router-Vlanif100] dhcp select global
[Router-Vlanif100] quit
[Router] interface vlanif 101
[Router-Vlanif101] ip address 10.23.101.1 24
[Router-Vlanif101] dhcp select global
[Router-Vlanif101] quit

4.5 步骤 4 配置 AP 上线

注意：仅给出 AC1 的配置过程，AC2 的配置参数跟 AC1 保持一致。
创建两个 AP 组“ap1”和“ap2”。
[AC1] wlan
[AC1-wlan-view] ap-group name ap1
[AC1-wlan-ap-group-ap1] quit
[AC1-wlan-view] ap-group name ap2
[AC1-wlan-ap-group-ap2] quit
创建域管理模板，在域管理模板下配置 AC 的国家码并在 AP 组下引用域管理模板。

[AC1-wlan-view] regulatory-domain-profile name default
[AC1-wlan-regulate-domain-default] country-code cn
[AC1-wlan-regulate-domain-default] quit
[AC1-wlan-view] ap-group name ap1
[AC1-wlan-ap-group-ap1] regulatory-domain-profile default
Warning: Modifying the country code will clear channel, power and antenna gain configurations of the radio and reset the AP. Continue?[Y/N]:y
[AC1-wlan-ap-group-ap1] quit
[AC1-wlan-view] ap-group name ap2
[AC1-wlan-ap-group-ap2] regulatory-domain-profile default
Warning: Modifying the country code will clear channel, power and antenna gain configurations of the radio and reset the AP. Continue?[Y/N]:y
[AC1-wlan-ap-group-ap2] quit
配置 AC 的源接口。
[AC1] capwap source interface vlanif 100
在 AC 上离线导入 AP1 和 AP2，并将 AP1 加入 AP 组"ap1"，AP2 加入 AP 组"ap2"。
[AC1-wlan-view] ap auth-mode mac-auth
[AC1-wlan-view] ap-id 0 ap-mac 00e0-fcf3-7340
[AC1-wlan-ap-0] ap-name area_1
[AC1-wlan-ap-0] ap-group ap1
Warning: This operation may cause AP reset. If the country code changes, it will clear channel, power and antenna gain configurations of the radio, Whether to continue? [Y/N]:y
[AC1-wlan-ap-0] quit
[AC1-wlan-view] ap-id 1 ap-mac 00e0-fc70-78d0
[AC1-wlan-ap-1] ap-name area_2
[AC1-wlan-ap-1] ap-group ap2
Warning: This operation may cause AP reset. If the country code changes, it will clear channel, power and antenna gain configurations of the radio, Whether to continue? [Y/N]:y
[AC1-wlan-ap-1] quit
将 AP 上电后，当执行命令 display ap all 查看到 AP 的"State"字段为"nor"时，表示 AP 正常上线。

4.6 步骤 5 配置 WLAN 业务参数

注意：仅给出AC1的配置过程，AC2 的配置参数跟AC1 保持一致。
创建名为"wlan-net"的安全模板，并配置安全策略。
[AC1-wlan-view] security-profile name wlan-net
[AC1-wlan-sec-prof-wlan-net] security wpa-wpa2 psk pass-phrase a1234567 aes
[AC1-wlan-sec-prof-wlan-net] quit
创建名为"wlan-net"的SSID 模板，并配置SSID 名称为"wlan-net"。
[AC1-wlan-view] ssid-profile name wlan-net
[AC1-wlan-ssid-prof-wlan-net] ssid wlan-net
[AC1-wlan-ssid-prof-wlan-net] quit
创建名为"wlan-net"的VAP 模板，配置业务数据转发模式、业务VLAN，并且引用安全模板和SSID 模板。
[AC1-wlan-view] vap-profile name wlan-net
[AC1-wlan-vap-prof-wlan-net] forward-mode direct-forward
[AC1-wlan-vap-prof-wlan-net] service-vlan vlan-id 101

[AC1-wlan-vap-prof-wlan-net] security-profile wlan-net
[AC1-wlan-vap-prof-wlan-net] ssid-profile wlan-net
[AC1-wlan-vap-prof-wlan-net] quit
配置AP 组引用VAP 模板，AP 上射频0 和射频1 都使用VAP 模板“wlan-net”的配置。
[AC1-wlan-view] ap-group name ap1
[AC1-wlan-ap-group-ap1] vap-profile wlan-net wlan 1 radio 0
[AC1-wlan-ap-group-ap1] vap-profile wlan-net wlan 1 radio 1
[AC1-wlan-ap-group-ap1] quit
[AC1-wlan-view] ap-group name ap2
[AC1-wlan-ap-group-ap2] vap-profile wlan-net wlan 1 radio 0
[AC1-wlan-ap-group-ap2] vap-profile wlan-net wlan 1 radio 1
[AC1-wlan-ap-group-ap2] quit

4.7 步骤 6 配置 AC1 和 AC2 负载分担方式的双链路备份功能

在AC1 上配置AC1 作为AP1 的主用AC、AP2 的备用AC，AC2 作为AP2 的主用AC、AP1 的备用AC。
[AC1-wlan-view] ac protect enable
Warning: This operation maybe cause AP reset, continue?[Y/N]:y
[AC1-wlan-view] ap-system-profile name ap-system1
[AC1-wlan-ap-system-prof-ap-system1] primary-access ip-address 10.23.100.2
[AC1-wlan-ap-system-prof-ap-system1] backup-access ip-address 10.23.100.3
[AC1-wlan-ap-system-prof-ap-system1] quit
[AC1-wlan-view] ap-system-profile name ap-system2
[AC1-wlan-ap-system-prof-ap-system2] primary-access ip-address 10.23.100.3
[AC1-wlan-ap-system-prof-ap-system2] backup-access ip-address 10.23.100.2
[AC1-wlan-ap-system-prof-ap-system2] quit
[AC1-wlan-view] ap-group name ap1
[AC1-wlan-ap-group-ap-group1] ap-system-profile ap-system1
Warning: This action may cause service interruption. Continue?[Y/N]y
[AC1-wlan-ap-group-ap-group1] quit
[AC1-wlan-view] ap-group name ap2
[AC1-wlan-ap-group-ap-group2] ap-system-profile ap-system2
Warning: This action may cause service interruption. Continue?[Y/N]y
[AC1-wlan-ap-group-ap-group2] quit
在AC2 上配置AC1 作为AP1 的主用AC、AP2 的备用AC，AC2 作为AP2 的主用AC、AP1 的备用AC。配置方式与AC1 上配置相同，请参考AC1 的配置。
在AC1 和AC2 上重启AP，下发双链路备份配置信息至AP。
[AC1-wlan-view] ap-reset all
Warning: Reset AP(s), continue?[Y/N]:y
[AC1-wlan-view] quit
[AC2-wlan-view] ap-reset all
Warning: Reset AP(s), continue?[Y/N]:y
[AC2-wlan-view] quit

4.8 步骤 7 配置双机热备份功能

在AC1 上创建HSB 主备服务0，并配置其主备通道IP 地址和端口号。
[AC1] hsb-service 0
[AC1-hsb-service-0] service-ip-port local-ip 10.23.102.1 peer-ip 10.23.102.2 local-data-port 10241 peer-data-port 10241

[AC1-hsb-service-0] quit

配置将WLAN 业务与NAC 业务绑定AC1 的HSB 主备服务。

[AC1] hsb-service-type ap hsb-service 0

[AC1] hsb-service-type access-user hsb-service 0

在AC2 上创建HSB 主备服务0，并配置其主备通道IP 地址和端口号。

[AC2] hsb-service 0

[AC2-hsb-service-0] service-ip-port local-ip 10.23.102.2 peer-ip 10.23.102.1 local-data-port 10241 peer-data-port 10241

[AC2-hsb-service-0] quit

配置将WLAN 业务与NAC 业务绑定AC2 的HSB 主备服务。

[AC2] hsb-service-type ap hsb-service 0

[AC2] hsb-service-type access-user hsb-service 0

4.9 步骤 8 检查配置结果

将AP 上电后，在AC1 上执行命令display ap all 查看到AP 的“State”字段为“nor”时，表示 AP 正常上线；AP 的“State”字段为“stdby”时，表示 AP 属于备用状态。

```
<AC1>display ap all
Info: This operation may take a few seconds. Please wait for a moment.done.
Total AP information:
nor  : normal          [1]
stdby: standby         [1]
--------------------------------------------------------------------------------
----------
ID   MAC            Name    Group  IP             Type            State  STA Uptime
--------------------------------------------------------------------------------
----------
0    00e0-fcf3-7340 area_1 ap1    10.23.100.253 AP4050DN-E      nor    0   17S
1    00e0-fc70-78d0 area_2 ap2    10.23.100.254 AP4050DN-E      stdby 0   -
--------------------------------------------------------------------------------
----------
Total: 2
```

```
<AC2>display ap all
Info: This operation may take a few seconds. Please wait for a moment.done.
Total AP information:
nor  : normal          [1]
stdby: standby         [1]
--------------------------------------------------------------------------------
----------
ID   MAC            Name    Group  IP             Type            State  STA Uptime
--------------------------------------------------------------------------------
----------
0    00e0-fcf3-7340 area_1 ap1    10.23.100.253 AP4050DN-E      stdby 0   -
1    00e0-fc70-78d0 area_2 ap2    10.23.100.254 AP4050DN-E      nor    0   1M:18S
--------------------------------------------------------------------------------
----------
Total: 2
<AC2>
```

两个 AC 中有一个掉线也不影响网络正常运行。（截图）